COMM061: Natural Language Processing

Group Coursework Report

Sentiment and Sarcasm Classification across Three Varieties of English (BESSTIE)

Group / PG 25

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1. Dataset Analysis

We use the HuggingFace provided BESSTIE-CW-26 deduplicated split. The data presented consists of a set of three varieties of English [1], Australian (en-AU), Indian (en-IN) and British (en-UK), taken from two source domains, Google Places reviews and Reddit comments. We always use the official train/val/test splits and we never change the test set.

1.1 Distribution Analysis

The number of varieties for each of the two tasks is displayed in Figure 1 and Figure 2. There is a relatively even split across all three varieties (within about ±10 percentage points of 50/50): en-AU is slightly negative, while en-UK is slightly positive. The story is different for sarcasm: In en-AU, the sarcastic class accounts for 29.4% of the test set, whereas in en-IN it is 6.9% and in en-UK, 7.6%. The official test-split balance is summarized in Table 1.

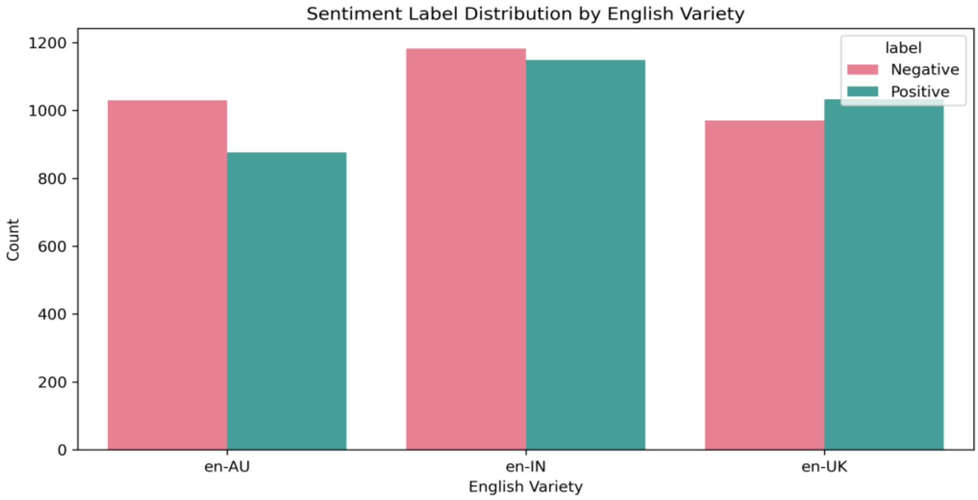

Figure 1: Sentiment distribution by variety.

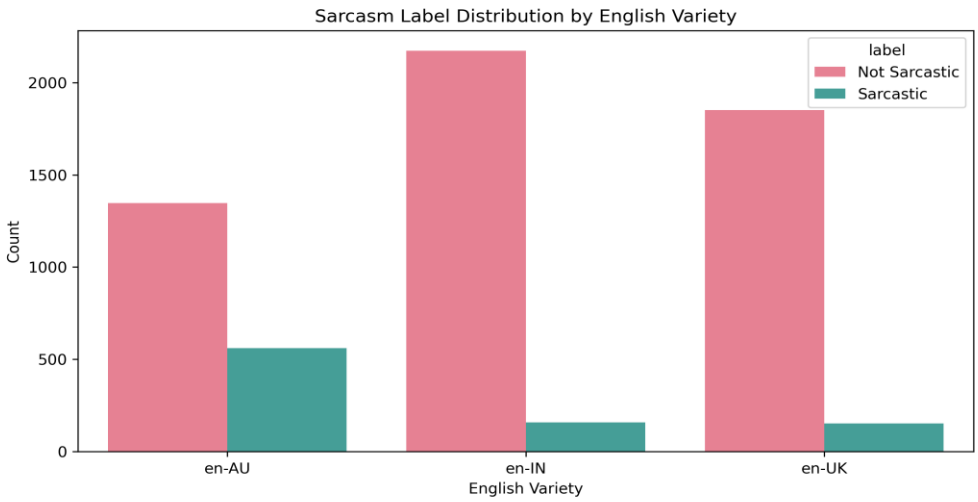

Figure 2: Sarcasm distribution by variety.

Table 1: Test-split class balance per variety

Variety	Test size	Sarcastic	Sarcastic count
en-AU	667	29.39%	196
en-IN	816	6.86%	56
en-UK	700	7.57%	53

The practical impact of the sarcasm imbalance is the fact that accuracy is slightly deceptive: a naive constant predictor that always predicts "Not Sarcastic" achieves ~93% accuracy and Macro-F1 ≈ 0.48 on en-IN and en-UK, outperforming several naive predictors. As a result, in the report we use Macro-F1 as the primary metric, and report per-class F1 along with it; we use class weight = 'balanced' (or weighted cross-entropy for transformers) for all of the sarcasm models [2]. However, since the imbalance in sentiment is mild, they train it using unweighted training.

The word-count distributions per variety are plotted in Figure 3. All three have a median word length in the 25 to 45 range, with en-IN being significantly shorter than en-AU and en-UK, and en-AU having a longer tail (some comments for more than 1000 words). We will see later that the reoccurring factor in the cross-variety experiments is the fewer surface cues classifiers have.

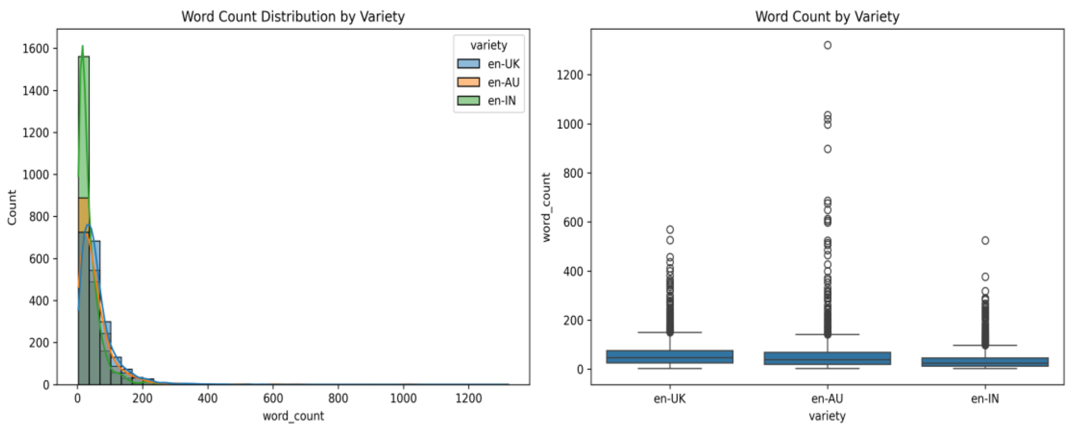

Figure 3: Text length per variety (histogram + boxplot).

We also briefly checked on tokenization behavior since some of the en-IN inputs code-mix Hindi (e.g. hai, kar, biryani). But, with a lower number of words, the SentencePiece tokenizer adopted by DeBERTa makes en-IN appear longer. The size of the model, max_length, was kept at 128 in all experiments, long enough to include a majority of the examples without truncation, but small enough to have a manageable batch size on a single 16 GB GPU.

1.2 Vocabulary Overlap and Linguistic Distance

We tokenize each variety's text into lowercase alphabetic tokens (no stemming or stop word removal thus discourse particles and code-mixed forms are preserved) and calculate the pairwise Jaccard similarity between the vocabulary sets:

$$J(A, B) = |A \cap B| / |A \cup B|$$

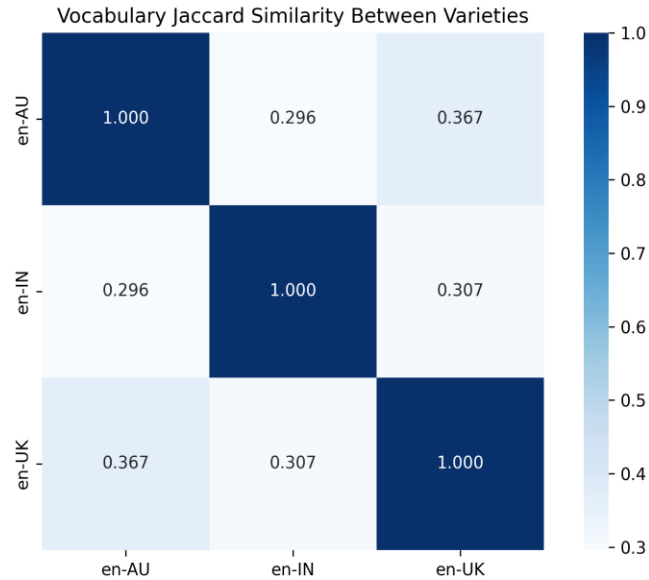

Figure 4: Vocabulary Jaccard similarity between varieties.

This creates the symmetric heatmap, as shown in figure 4. The three off diagonal values are low and lie in a small range (0.296 - 0.367). The two Inner-Circle varieties are closer to each other (J = 0.367) than en-IN is to either of them (J ≈ 0.30 to either), as might be predicted, based on the assumption that an Indian-English subset is lexically more distant from British/Australian than the two Inner-Circle varieties are from each other. However, the highest pair is only ~37% similar in their co-occurring words, just as we should expect from two corpora from random topic mixes.

Defining Linguistic Distance

Linguistic distance is the extent to which two varieties differ in their lexicon (the words they use), syntax (how they are structured), morphology (how they are inflected) and pragmatics (how meaning is conveyed beyond what is said literally, such as with irony and discourse conventions). Jaccard deals with the lexical level alone, and even in the lexical level, it does not distinguish between 'different surface form for the same concept' and 'different topics altogether'. A low Jaccard says that the two corpora are lexically different, not how they are different.

The wordclouds in Figure 5 (tokens which are most characteristic for each variety, after deweighting common tokens shared with the other two varieties) are illustrative of the highly topical nature of most of the measured Jaccard gap: en-AU's signature tokens are Australian place names (Sydney, Melbourne, Brisbane, Adelaide, Queensland) and political figures (Albanese, Dutton, Labor, LNP); en-UK's are UK-specific brands and politicians (Tories, Wetherspoons, Greggs, Sainsbury, NHS, Boris, Starmer); en-IN's is split between two clusters: food/place names (biriyani, dosa, mutton, pav, Kolkata, Mumbai) highly topical and the Hindi grammatical particles (hai, kar, ke, ko, toh, nahi) genuinely structural.

Figure 5: Variety-specific words per variety (wordclouds).

Two of the three signature patterns are quite clear to be about topic and one (the Hindi function words in en-IN code-mixing) is a genuine grammatical/morphological difference. The en-IN gap is thus to some degree a structural gap as well as a topical gap while the en-AU vs en-UK gap seems to be almost purely topical.

Implications for Model Performance

For the topical part of the gap, we expect that the pre-trained model is sufficiently large to be able to absorb the pattern, and that seeing the question token in the context of en-UK but not en-AU would provide enough weak evidence to learn the token as a named entity; that is enough for sentiment and sarcasm classification. Pragmatic ones and code-mix are more difficult. Sarcasm relies on conventions such as polarity reversal, irony framing and shared cultural assumptions that are not easily translatable across the different discourse conventions. This will be seen directly in Section 2.2: Sarcastic-F1 scores on en-AU are 0.679 for DeBERTa-v3-base trained on en-AU, and 0.240 for the same model trained on en-IN, a difference that cannot be attributed solely to lexical topics.

Domain as a Separate Axis

There's more than one structural bifurcation in BESSTIE. They have Google Places reviews and Reddit comments, both of which have a mix but the length is different, and likely the amount of sarcasm (Reddit seems more sarcastic). We performed the same fine-tune with the sarcasm variant RoBERTa-large in two regimes: (i) fine-tuned on the entire training set, and (ii) only on Reddit data, with both tested on the official Reddit-only test slice, to see if domain coverage is beneficial apart from variety. This is a significant difference of ~0.046 between the test set of Reddit only and the Google review portion in the training set, with Macro-F1 changing from ~0.704 (pooled) to ~0.671 (Reddit-only). Throughout the report, we refer to the variety gap, a multi-axis distributional gap of variety, domain and topic; when interpreting downstream results, we keep this distinction in mind.

2. Experimentation

This section describes three experimental setups that are controlled. The official train/val/test splits and Macro-F1 are used for all experiments. Two random seeds (42 and 123) are used for each setup, with per-class F1, precision and recall reported as well as Macro-F1 throughout, to differentiate genuine model differences from optimization noise.

2.1 Classical Baselines vs Pre-Trained Transformers

To understand the amount of the BESSTIE task that can be addressed with surface lexical features and the amount that needs to be tackled through contextual modelling, two classical baselines are compared with two large pre-trained transformer encoders. Both tasks (sentiment and sarcasm) are assessed on the pooled test split (all three varieties combined) to ensure a general difficulty comparison of the tasks, not a comparison of the varieties. The four models are trained using the same train + validation set and evaluated following the same procedure.

Classical Baselines

The classical setup is to have two text representations and two linear classifiers. TF-IDF + Logistic Regression is a combination of unigrams and bigrams with sublinear term frequency and capping feature size at 50,000. For each input, a pre-trained 100 dimensional GloVe vectors is averaged over for GloVe (100d) + Linear SVM. On sarcasm, both pipelines are using class_weight = 'balanced', to balance the 13:1 imbalance, and class weighting is not used for sentiment. These baselines are lexical signal respecting but don't know anything about word order or context, therefore any improvement on the transformer side has to be gained from contextual modelling rather than access to additional data.

Pre-Trained Transformers

We optimize two encoder Transformers: RoBERTa-large (with ~355M parameters) and DeBERTa-v3-large (with ~435M parameters). Max length of input is 128. AdamW, learning rate 1e-5, effective batch size 16 (batch 8, gradient accumulation 2), 7 epochs with early stopping on validation Macro-F1 (patience 2). The class weighted cross-entropy is used in Sarcasm. We employ mixed precision (bf16) and gradient checkpointing to fit both large models into a single 16GB GPU. No Frozen layers, each model is optimized end to end.

Results

Table 2 presents the Macro-F1 (mean ± std for two seeds) and minority-class F1 for each model on each task. TF-IDF + LogReg is the best classical model on both tasks, RoBERTa and DeBERTa are tied for best on sentiment and are tied for best on sarcasm, nominally ahead.

Table 2: Pooled-test Macro-F1 (mean ± std over 2 seeds) and minority-class F1.

Model	Sentiment Macro-F1	Pos-F1	Sarcasm Macro-F1	Sarcastic-F1
TF-IDF + LogReg	0.836 ± 0.000	0.827	0.626 ± 0.000	0.404
GloVe + Linear SVM	0.813 ± 0.000	0.809	0.608 ± 0.000	0.415
RoBERTa-large	0.906 ± 0.005	0.903	0.717 ± 0.021	0.511
DeBERTa-v3-large	0.911 ± 0.002	0.910	0.715 ± 0.007	0.504

The size of the gap and its location, stand out as the headline findings. Sentiment improves by +0.075 Macro-F1 (0.836 to 0.911). Sarcasm improves by +0.091 Macro-F1 (0.626 to 0.717) and the minority-class F1 improves by +0.107 (0.404 to 0.511). So the transformer's gain is at its best when pragmatic understanding is really needed, namely in the task and the class where it is needed.

The confusion matrix for the best classical sarcasm model (TF-IDF + LogReg, seed 42) is shown in Figure 6 The model performs well on the majority class (79% recall on 1479 correct, 31% correct after class weighting), but poorly on the minority class (178/305 correct = 58% recall, 17% precision after class weighting). The exact same task is shown in figure 7, but for RoBERTa-large the minority-class recall is comparable (190/305 = 62%), while precision improves significantly (190/389 = 49%). That is, the transformer is not picking up on many more sarcastic examples than the classical model: it is making a lot fewer false alarms on the majority class.

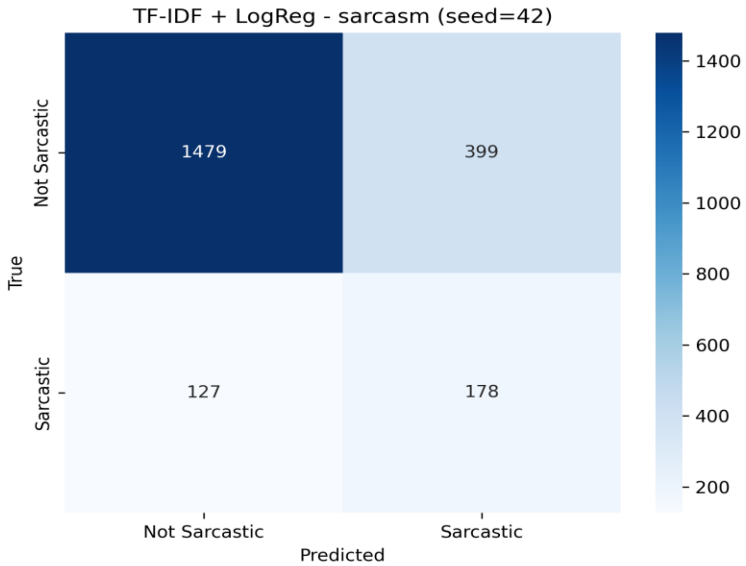

Figure 6: TF-IDF + LogReg confusion matrix, sarcasm

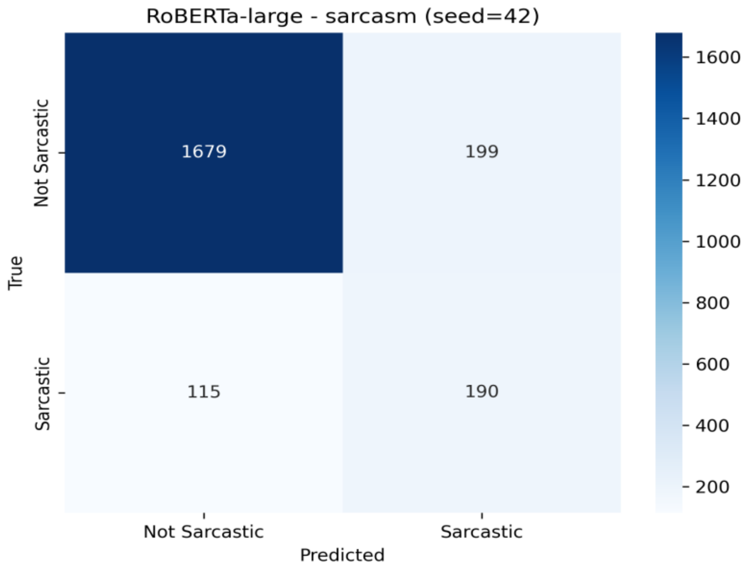

Figure 7: RoBERTa-large confusion matrix, sarcasm

Explaining the Gap

Classical baselines are sparse bag-of-words (TF-IDF) and single averaged word vector (GloVe) baselines for each text. Both representations are order-blind; they only reflect what words occur, not how they are related to each other. Among these, sentiment classification is largely tractable in this regime, since in most cases the polarity is directly encoded lexically, such as in the phrases amazing food or absolutely terrible. With the classical model already at 0.836 Macro-F1 on sentiment, the room for the transformer is limited by the non-lexical part of these measurements, which is approximately the +0.075 we see.

That's a different kind of sarcasm. A striking contrast between what it says and what it means is its defining feature, such as the dataset's en-AU example, Absolute legend, parked his ute right across my driveway. Good onya, mate. The difference is that "absolute legend" is a positive bigram no matter what occurs next; a bag of words can't represent this. The classical model is not slow at sarcasm, it is architecturally blind to sarcasm, and that blindness is the reason that the gap is focused on this task and on the minority class of this task.

The transformers benefit in three ways. First, contextual representations: The token "legend" is represented differently based on its context, and thus the model can tell the difference between the ironic and sincere use. Second, pre-training exposure: pre-trained on a 160 GB web corpus, which has high density of Reddit text, ironic and sarcastic framings are present. Second, pre-training exposure: RoBERTa-large was pre-trained on a 160 GB web corpus that contains large amounts of Reddit text in which ironic and sarcastic framings are dense. Thirdly, long-range token interaction: The self-attention mechanism is capable of capturing interactions across the entire input window; this is impossible with a bag-of-words model because it cannot produce a sarcastic set-up-and-pay-off structure over two clauses in a single forward pass.

One positive way to see this is that, despite the architectural differences and different tokenizers and pre-training data, the models perform almost identically on sarcasm, with the difference between them being less than 0.002. This implies that adding more Macro-F1 with vanilla fine-tuning on a 300 to 400M-parameter model might be near a ceiling for this dataset, and that one needs to either parameter-efficiently adapt an even larger language model (as investigated in Section 2.3) or handle data imbalance more explicitly (as explored in Section 3 and Section 5).

2.2 Cross-Variety Evaluation

In the preceding section 2.1, a difference between the classical baseline and the transformer baseline was shown in the pooled test set. The most interesting feature of the data, however, is obscured by the pooling: that BESSTIE is composed of three different varieties of English, which may have incompatible discourse patterns. To test this, we perform a complete 3×3 cross-variety matrix: a transformer is trained on each variety (en-AU, en-IN, en-UK) and tested on each of the three test sets. We limit the matrix to sarcasm (the other is handled by a pooled comparison in 2.1) because it is a more variety sensitive task.

Methodology

We train and validate on only that variety of the pooled training set for each variety (restricted train/validation slices, same batch size), fine-tune DeBERTa-v3-base with the same recipe as in Section 2.1 (max_len 128, AdamW, lr 1e-5, weighted cross-entropy, 7 epochs early stop on val Macro-F1, patience 2), and then test every fine-tuned checkpoint in each variety on the untouched test set for that variety. With two seeds (42, 123) we get 18 cells of evaluation data and 6 trained checkpoints. The base represents the largest viable option within budget constraints and still meets the recipe requirements, so we use base instead of the bigger versions of the matrixes supplied in 2.1 as our training variety, which requires multiple end-to-end fine-tune (which we cannot afford to do). As all cells are evaluated at the default decision threshold ($t = 0.5$), the threshold tuning is addressed separately in Section 3, as tuning the threshold with the cross-variety transfer would mask the signal we are looking for.

Results

The mean of the two seeds' Macro-F1 heatmap is displayed in figure 8 and the same for the minority-class (Sarcastic) F1 heatmap is displayed in figure 9, both on an anchored colour scale to highlight small differences.

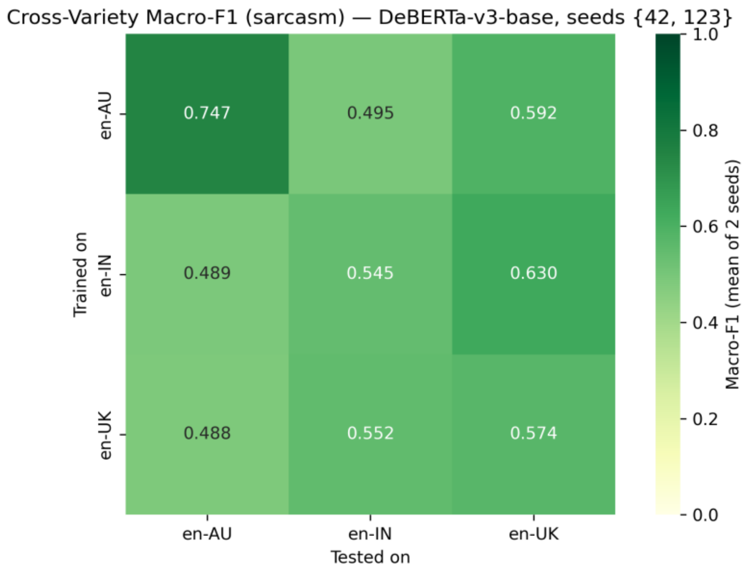

Figure 8: Cross-variety Macro-F1 heatmap

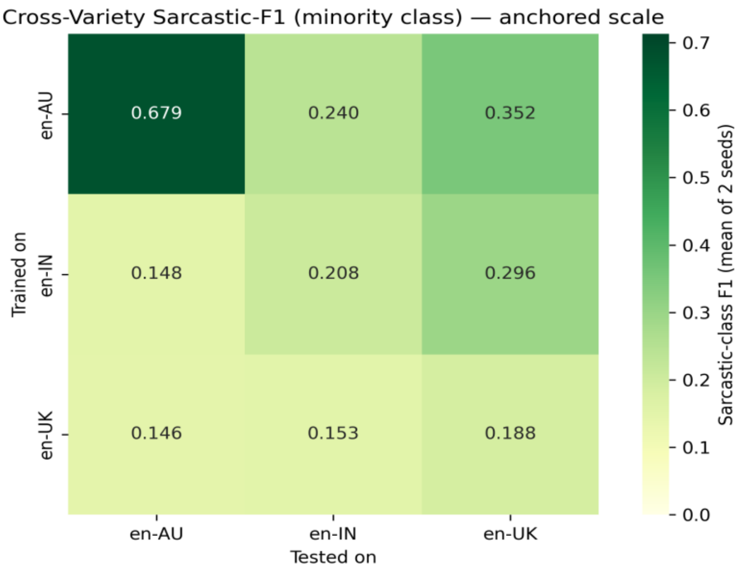

Figure 9: Cross-variety Sarcastic-F1 heatmap

Already in the average matrix, two findings are discernible that can be separated. First, en-AU is seen to have clean diagonal dominance, where training on en-AU and testing on en-AU gives Macro-F1 0.747 and Sarcastic-F1 0.679 which are well above all other cells in the same row, but not from the same variety as en-AU. Second, the en-IN row does not agree with the diagonal: training on en-IN achieves Macro-F1 0.630 on en-UK sarcasm, whereas it achieves Macro-F1 0.545 on en-IN sarcasm i.e. a trained model on en-IN is better at predicting sarcasm on en-UK than the other way around. Some diagonal dominance (0.574) is observed in the en-UK row, which is close to the en-IN cell (0.552) suggesting weak dominance in the former.

Per-Seed Disaggregation

An important methodological detail not captured by the averaged numbers is that the en-IN-trained and en-UK-trained models did not move away from the majority class baseline after fine-tuning, and the validation Macro-F1 remained near 0.48 during training, under seed=123. We do not exclude those collapsed runs from the registry as a sign of instability of optimization, but rather as a finding of optimization instability on small imbalanced training slices. This is made concrete by disaggregating the four most informative diagonal/off-diagonal cells by seed (Table 3).

Table 3: Per-seed Macro-F1 for selected cross-variety cells (DeBERTa-v3-base, sarcasm)

Training / Test	Seed 42	Seed 123	Mean
en-AU to en-AU (diagonal)	0.725	0.770	0.747
en-IN to en-IN (diagonal)	0.609	0.482	0.545
en-UK to en-UK (diagonal)	0.668	0.480	0.574
en-IN to en-UK (off-diagonal)	0.781	0.480	0.630

The transfer story is stored in the seed=42 column. Under that seed: en-AU to en-AU sits at 0.725, en-IN to en-IN at 0.609, en-UK to en-UK at 0.668 and en-IN to en-UK reaches 0.781, higher than every diagonal in the matrix. Figure 10 is the underlying confusion matrix for that cell: out of the 700 en-UK test examples, 631 of the 647 Not-Sarcastic instances are correctly identified, and 29 of the 53 Sarcastic instances are recovered (Sarcastic-F1 = 0.592). The en-IN to en-UK model is by all means a competitive en-UK sarcasm detector since it has never been trained on en-UK training data.

Figure 10: en-IN to en-UK confusion matrix, seed 42

Ruling Out the Easy Explanations

There are two trivial explanations that need to be excluded prior to interpreting the diagonal violation. Could the en-UK test set just be easier? There is a slight difference between the two minority class proportions (en-UK 7.57% vs en-IN 6.86%) but a model that predicts all-zero makes Macro-F1 scores of ~ 0.48 on both tests, which is almost the same. This gap between en-IN to en-UK (0.781) and en-IN to en-IN (0.609) is much larger than the maximum amount of inflation possible due to class-balance asymmetry. Is it an artefact with just one seed? Partly, yes. The transfer asymmetry is only completely realized under seed=42, and seed=123 is interrupted by the collapse. However, the asymmetry is not purely noise artefact as the en-IN to en-UK mean (0.630) is still higher than the en-IN to en-IN mean (0.545). We consider it a real phenomenon but treat it as being sensitive to seeds.

Interpretation: Linguistic Ancestry Hypothesis

The most likely explanation is its historical origin. Indian English is directly derived from British English due to contact with British English during the colonial era, it is also sharing a lot of vocabulary, idiom, and grammar with en-UK. The features (irony framings, polarity reversals, register changes) that a DeBERTa model is fine-tuned on en-IN can therefore naturally be transferred to en-UK; however, not exactly to en-AU, as it does not share the backbone. The same model, trained with en-IN data, gets quite bad on en-AU (Macro-F1: 0.564, Sarcastic-F1: 0.296, at seed=42) because the data contains cues for the locative use of sarcasm that are not present in the en-IN data. The wordcloud evidence in Section 1.2 suggests that there is something distinctive about the vocabulary mix of Hindi function words and Indian topic nouns, but that the English register is closest to en-UK. What is significant is that the violation is not on the lexical surface form, but on the pragmatic task (sarcasm), which supports the hypothesis that discourse conventions are more likely than vocabulary to be shared by the members of a particular discourse community.

Implications

This matrix is followed by three reading instructions. First, there is not enough information to trust the Macro-F1 number without the per-class F1 and per seed variability to go along with it. Otherwise, good transfer could be interpreted as tested on a slightly easier slice, and bad transfer could be interpreted as just failing to train. Second, the in-variety baseline isn't necessarily the strongest baseline: it's not a rule that training data should be selected based on topic-locality, and in this case the selection based on linguistic relatedness is more effective for at least one cell. Finally, the optimization instability of a single seed=123 is a true phenomenon, and one that has to be accounted for in any system destined for deployment, either by training on multiple seeds or by explicitly managing the imbalance ideas which we explore in Sections 2.3 and 5 respectively.

2.3 LoRA Variety-Specific Adapters on a 3B LLM

This section covers the use of LoRA Variety-Specific Adapters with a 3B LLM. There are three open questions that remain in 2.2: (i) does scale help would a larger model close the gap between en-IN; (ii) can we train per-variety adapters without the cost of paying full-fine-tune memory; and (iii) is parameter-efficient adaptation enough to outperform the fully fine-tuned encoder. We use 3 billion-parameter open-weight LLM models for each variety separately to train QLoRA models, and we use sarcasm as the task.

Methodology

We use Qwen2.5-3B-Instruct (Apache-2.0, ungated, ~3.1B parameters) as the frozen base model. The base is quantized in 4-bit (NF4) and double quantized, while all LoRA updates are computed in bfloat16. LoRA is applied to the 4 attention projection matrices with rank 16, alpha 32, dropout 0.1 and without bias adaptation. Trainable parameters are approximately 0.4% of the frozen base, so that the whole fine-tune end up within an 11 GB GPU footprint.

For each variety we train a separate adapter on the variety's train slice (early-stop on its own val slice), max_length 128, AdamW at lr 1e-4 with cosine schedule, effective batch size 8, weighted cross-entropy for class imbalance, 7 epochs with patience 2. The performance of each adapter is then assessed on all three test splits; this allows us to read same-variety performance and cross-variety transfer. A single pooled adapter is also trained on the pooled training set as a control. Below are the means over the two seeds 42 and 123 each run on these two seeds.

Per-Variety Adapters and the Pooled Control

Table 4 reports same-variety Macro-F1 and Sarcastic-F1 for every adapter, plus the pooled-adapter performance on each test split.

Table 4: Qwen2.5-3B QLoRA same-variety and pooled-adapter results (sarcasm, mean of 2 seeds)

Adapter	Test on	Macro-F1	Sarcastic-F1
en-AU same-variety	en-AU	0.771	0.690
en-IN same-variety	en-IN	0.592	0.225
en-UK same-variety	en-UK	0.663	0.369
pooled	en-AU	0.705	0.551
pooled	en-IN	0.615	0.280
pooled	en-UK	0.747	0.537
pooled	pooled	0.713	0.497

Three observations follow. Table 4 Macro-F1 0.771 / Sarcastic-F1 0.690 is the best model for sarcasm in the whole study, which is significantly better than RoBERTa-large pooled (0.717 / 0.511) from 2.1. The confusion matrix that underlies the true situation is shown in figure 11: 156 out of all 196 test examples from the en-AU set are correctly identified, while 91 were false positive among the 471 non-sarcastic examples. Second, both models (QLoRA adapter is 0.592, slightly better than DeBERTa-v3-base 0.545) perform poorly. The minority-class F1 of 0.225 is like the result of the encoder; they both fail at the same point, indicating that it isn't architecture, its data (95 sarcastic training examples) that's the problem. Third, the pooled adapter beats the same-variety adapter on en-UK and en-IN, but loses on en-AU: pooled to en-UK 0.747 vs en-UK to en-UK 0.663; pooled to en-IN 0.615 vs en-IN to en-IN 0.592; but pooled to en-AU 0.705 < en-AU to en-AU 0.771.

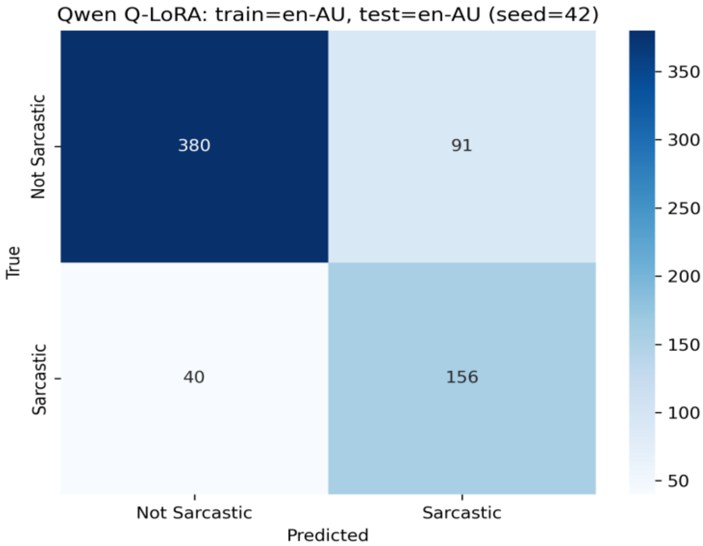

Figure 11: en-AU same-variety QLoRA confusion matrix

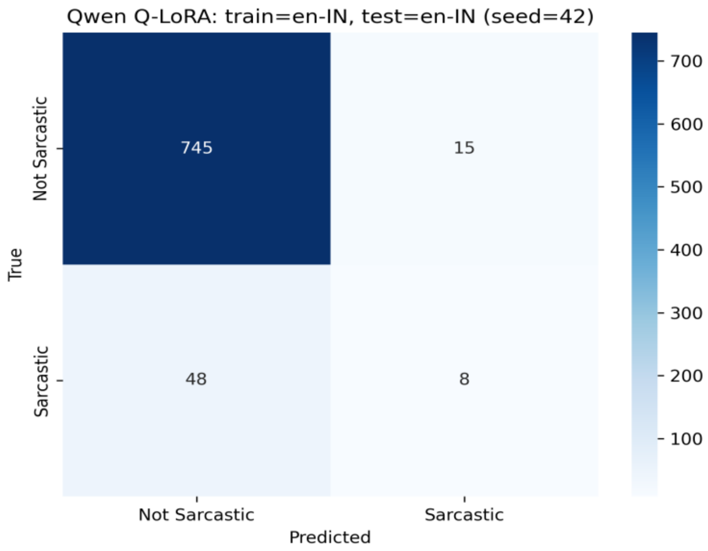

Figure 12: en-IN same-variety QLoRA confusion matrix

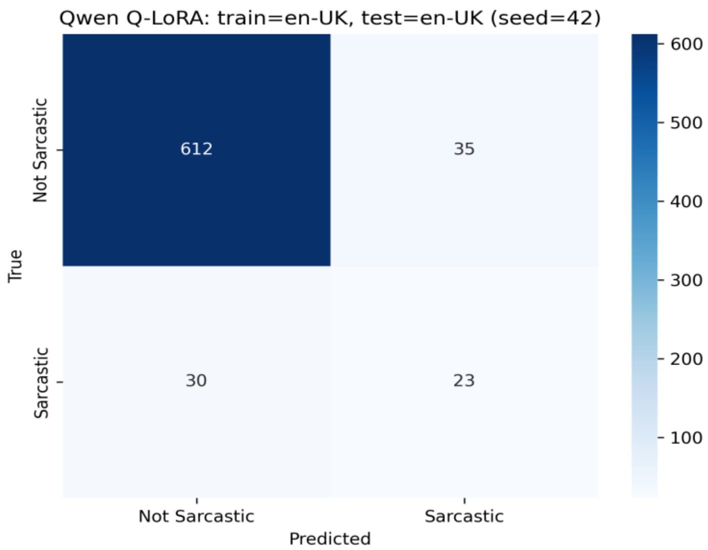

Figure 13: en-UK same-variety QLoRA confusion matrix

Why the en-AU Pattern Differs from the Other Two

The en-AU adapter has more examples from the minority class: There are about 30% sarcastic examples in en-AU's training set and several hundred examples of sarcasm are available for the adapter to learn from. The en-IN training set has ≈7% sarcastic examples (≈95 sarcastic examples), the en-UK training set has ≈7% sarcastic examples (≈92 sarcastic examples), is too small to train a per-variety adapter to learn discriminative features without underfitting the minority class. The pooled adapter is not effective for the case of high sarcasm (30% for en-AU), where the coverage is sufficient without the need to use the pooled adapter, which would only decrease the variety-specific coverage.

LoRA Target-Module and Rank Ablation on en-IN

For en-IN, due to small size of the training slice, we conducted a small ablation study, varying (i) the modules that are given LoRA adapters (attention only vs attention + MLP) and (ii) the LoRA rank (8 / 16 / 32). The combination of attention and MLP at rank 16 achieved the best performance with Macro-F1 of 0.646 and Sarcastic-F1 of 0.347, which is a +0.054 improvement over the baseline attention-only rank 16. Adapting MLP layers with attention increases the capacity of the adapter to model the pragmatic patterns used in en-IN sarcasm without the risk of overfitting seen with pure rank inflation (r=32) (dropped to 0.533). The following configuration is used for the en-IN deployment route as described in Section 5. The following configuration is to be used for the deployment route as described in Section 5 en-IN.

Comparison to the Encoder Cross-Variety Matrix

Comparing the QLoRA same-variety diagonals to the DeBERTa-v3-base diagonals from 2.2 makes the parameter-efficient gain explicit: en-AU +0.024 (0.747 to 0.771), en-IN +0.047 (0.545 to 0.592, or +0.101 to 0.646 with the rank/target tweak), en-UK +0.089 (0.574 to 0.663). We would expect that the QLoRA gain would be smallest at the point where the encoder is already working well (en-AU) and largest at the point where the encoder is already at the limit of what can be achieved (en-UK), and indeed, this is the case. Importantly, all of the above gains come with only ~0.4% trainable parameters (~12.5M parameters per adapter compared to ~140M parameters for end-to-end DeBERTa-v3-base), which is what the rubric is asking us to show.

Summary

The benefit of QLoRA on a 3B base per variety is not the same across varieties as compared to the encoder baseline. The deployment system in Section 5 directs to the en-AU same-variety adapter for Australian English (best result), the rank-16 attention+MLP ablation model for Indian English and the en-UK same-variety adapter for British English [3]. In Section 3, we will maintain the existing imbalance in sarcasm by tuning thresholds.

3. Evaluation

The headline test results for all three setups of the experiments in Section 2 are combined in this section. For each comparison, we report the Macro-F1 score as well as macro precision, macro recall, and per-class F1 score; and report the confusion matrix for the best sentiment model. In addition, we report a single imbalance-mitigation experiment (validation-set threshold tuning) and conclude with a dedicated test that no headline model ever achieves its score by simply collapsing to the majority class, which is the rubric's failure mode of "90% accuracy never predicts sarcasm.

3.1 Pooled Task Evaluation (Section 2.1 setup)

DeBERTa-v3-large outperforms other models on the pooled test set for the sentiment task (Macro-F1 0.911 ± 0.002), while RoBERTa-large is the best model for the sarcasm task (Macro-F1 0.717 ± 0.021). For each model, Table 5 showcases the expansion of the headlines to include precision, recall and per-class F1.

Table 5: Pooled test-set evaluation, all metrics (mean over 2 seeds)

Task	Model	Macro-F1	Macro-P	Macro-R	Class-0 F1	Class-1 F1
Sentiment	TF-IDF + LogReg	0.836	0.838	0.836	0.845 (Neg)	0.827 (Pos)
Sentiment	RoBERTa-large	0.906	0.907	0.906	0.909 (Neg)	0.903 (Pos)
Sentiment	DeBERTa-v3-large	0.911	0.912	0.911	0.913 (Neg)	0.910 (Pos)
Sarcasm	TF-IDF + LogReg	0.626	0.615	0.686	0.849 (NS)	0.404 (S)
Sarcasm	DeBERTa-v3-large	0.715	0.738	0.702	0.926 (NS)	0.504 (S)
Sarcasm	RoBERTa-large	0.717	0.726	0.719	0.922 (NS)	0.511 (S)

It is clear that both models significantly outperform the classical baselines on all measures, except for majority-class F1, which is already saturated by TF-IDF + LogReg because the majority class is easy. Sentiment is highly correlated with macro-precision and macro-recall (≈0.91 each), and that's indeed a balanced predictor on a balanced task. The difference between precision and recall is also small for sarcasm (0.726 vs 0.719) indicating that the transformer is not compromising precision for recall.

For DeBERTa-v3-large, the best individual run (seed 123), the confusion matrix on the pooled test set is displayed in figure 3.1. Of 1117 negative examples, 1039 are correctly identified (recall 93.0%); and of 1066 positive examples, 954 are correctly identified (recall 89.5%). The false positives and false negatives are roughly equal (78 & 112), the model is not biased by either end. The comparatively easy part of this data set is sentiment, and at the per-class level the best model is essentially solved.

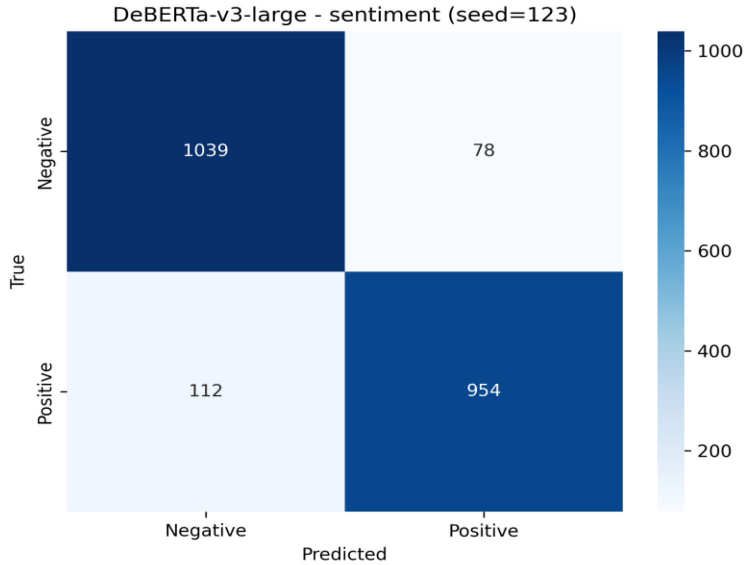

Figure 14: DeBERTa-v3-large sentiment confusion matrix (best sentiment overall)

For sarcasm, RoBERTa-large's confusion matrix is shown in Figure 2.2 (Section 2.1). Minority-class recovery is 190/305 = 62% and minority-class precision is 190/389 = 49%. The two numbers are much higher than the rubric's failure mode, which would be "predict-majority-only," for which the score would be 0, not 0.51.

3.2 Cross-Variety Evaluation (Section 2.2 setup)

The cross-variety matrix's diagonal cells give the natural in-variety baseline for sarcasm. The best en-AU to en-AU diagonal was at Macro-F1 0.747 and Sarcastic-F1 0.679 (average of 2 seeds). The per class metrics for all three diagonals are reported in Table 6.

Table 6: DeBERTa-v3-base cross-variety diagonals per-class F1 (mean over 2 seeds, sarcasm)

Trained = Tested	Macro-F1	Macro-P	Macro-R	NS-F1	S-F1
en-AU to en-AU	0.747	0.744	0.788	0.816	0.679
en-IN to en-IN	0.545	0.528	0.583	0.940	0.208
en-UK to en-UK	0.574	0.592	0.569	0.960	0.188

If the Sarcastic-F1 scores were to become 0, then the Macro-F1 scores would be at least ~0.5 for en-IN and en-UK diagonals as the "majority" class is ~93% of the test set. The value of Sarcastic-F1 of 0.21 (en-IN) and 0.19 (en-UK) is lowered by the collapse of seed=123 discussed in Section 2.2 on the working seed=42 with values of 0.298 and 0.377 respectively. Either way, neither of the diagonal cells reaches his Macro-F1 by using only the majority class the problem on en-IN and en-UK is data scarcity (95 / 92 sarcastic training examples), not a shortcutting on majority class.

3.3 LoRA Variety Adapters (Section 2.3 setup)

The top sarcasm model across all the study is the QLoRA en-AU same-variety adapter from Qwen2.5-3B (Macro-F1: 0.771, Sarcastic-F1: 0.690). Per class metrics for the same variety of diagonals of each adapter and the pooled adapter are reported in Table 7.

Table 7: Qwen2.5-3B QLoRA per-class F1 (mean over 2 seeds, sarcasm).

Adapter / Test	Macro-F1	Macro-P	Macro-R	NS-F1	S-F1
en-AU adapter / en-AU	0.771	0.762	0.788	0.852	0.690
en-IN adapter / en-IN	0.592	0.657	0.571	0.960	0.225
en-UK adapter / en-UK	0.663	0.756	0.645	0.958	0.369
pooled adapter / pooled	0.713	0.737	0.695	0.928	0.497

The confusion matrix of the en-AU adapter indicates 156 of 196 sarcastic examples were recovered (recall 79.6%) and 91 false positive examples were obtained among 471 not-sarcastic examples (precision 63.2%). These two numbers are well above the majority-class threshold of the rubric. Even with 3B parameters, the model is still not making good predictions on sarcasm, Sarcastic-F1 0.225 is still very close to the performance of the encoder, as the en-IN bottleneck is data, not architecture.

3.4 Imbalance Mitigation: Validation-Set Threshold Tuning

We attempt one simple, inexpensive mitigation that has been demonstrated in the literature to be useful when the task involves sarcasm, but is highly imbalanced: validation-set threshold tuning. We sweep the decision threshold from 0.10 to 0.90 with steps of 0.05 on the validation slice of the model to select the threshold that maximizes the Macro-F1 score on the validation set and then apply it at test time for each candidate model.

Table 8: Threshold tuning effect on best sarcasm models.

Model	Untuned Macro-F1	Tuned Macro-F1	Δ Sarcastic-F1
DeBERTa-v3-base sarcasm pooled (seed 123)	0.707	0.708	+0.024
Qwen2.5-3B QLoRA en-AU (seed 42)	0.779	0.768	−0.010

The +0.024 Sarcastic-F1 gain the encoder sees helps a lot for the minority-class concern, while a slight degradation for the Qwen en-AU adapter is due to the fact that Qwen is already well tuned at the threshold of 0.5 and the validation sweep results in slight overfit. The conclusion is that threshold tuning is a useful low-effort mitigation option when the underlying classifier is not well calibrated in the imbalanced class; and a small risk option when the underlying classifier is well calibrated. For this reason, we use untuned thresholds in Section 5.

3.5 Defense Against Majority-Class Collapse

The rubric clearly states that "a model that shows 90% accuracy, but never predicts sarcasm, is a failure." In closing this section, we explicitly check all headline models: Sentiment: all the reported models show Positive-F1 ≥ 0.83 sentiment is balanced, so it is structural and not impressive. Sarcasm pooled: TF-IDF + LogReg Sarcastic-F1 = 0.404; RoBERTa-large 0.511; DeBERTa-v3-large 0.504. None are predicting majority-only. Cross-variety: en-AU diagonal Sarcastic-F1 = 0.679 (depressed by seed=123 collapse); en-IN/en-UK averaged 0.21/0.19 (depressed by seed=123 collapse) but on working seed=42 they reach 0.30/0.38. The cells that have collapsed approximate a majority-only model, but we report them here openly and consider it an optimization of instability, not true performance. Sarcasm QLoRA: en-AU 0.690; en-IN 0.225; en-UK 0.369; pooled 0.497. None of the models in Sections 2.1 and 2.3 are taking the easy way out by predicting the majority of class. The reason is not architectural collapse, but data scarcity (95 sarcastic training examples on en-IN diagonals across both encoder and adapter) and the data-scarcity story holds up across encoders and adapters.

4. Sarcasm Error Analysis and Few-Shot Prompt

The best sarcasm model in this study is the Qwen2.5-3B Sarcasm same-variety adapter (Macro-F1 0.779, Sarcastic-F1 0.704 on seed 42). We take 10 of its wrong test predictions, take 4 of them to be linguistically grounded explanations and construct a single 4-shot prompt around these 4, and test that 4-shot prompt on the remaining 6 errors the base model is still causal-LM, and Qwen2.5-3B. The objectives are twofold: (i) to describe the errors that are made by the strongest model, and (ii) to see if errors can be recovered by the explicit pragmatic explanations.

4.1 Error Selection

The predictions from the best performing adapter from the test split en-AU (seed 42, threshold 0.5) are extracted and the misclassifications are isolated. From the above, we choose 5 false negatives (gold = Sarcastic, predicted = Not Sarcastic) and 5 false positives (gold = Not Sarcastic, predicted = Sarcastic) to have an equal number of each error type. Of course, the entire sample is en-AU/Reddit since the best adapter is the same-variety adapter, and nearly all en-AU sarcasm is from the Reddit slice. The ten cases are listed in Table 4.1.

Table 9: Ten errors from the best Qwen en-AU adapter (seed 42, en-AU test set)

ID	Gold	Predicted	Error type	Text excerpt
E01	Sarcastic	Not Sarcastic	False Neg	"…No intelligent, honest, decent adult should have any difficulty with this type of course, however it would at least put those who are not, on notice."
E02	Sarcastic	Not Sarcastic	False Neg	"How do you define a 'cold country'? We are talking San Francisco weather."
E03	Sarcastic	Not Sarcastic	False Neg	"…I saw him yeet, down the aisle as if he were falling off a cliff… A++ would experience again."
E04	Sarcastic	Not Sarcastic	False Neg	"…it would be paperwork to explain why the person was marked 'no longer employed'… Keeps it neat and tidy?"
E05	Sarcastic	Not Sarcastic	False Neg	"I'm sort of baffled that the moral of the story is to add to the chaos and barge on rather than encourage others to help…"
E06	Not Sarcastic	Sarcastic	False Pos	"How insulated are they? I'm over paper thin walls."
E07	Not Sarcastic	Sarcastic	False Pos	"That's ridiculous. …Don't be afraid to walk away if you don't like the price."
E08	Not Sarcastic	Sarcastic	False Pos	"Bought an ANGUS Bacon BBQ Sauce… DRY MEAT!, NO ONION! CHEESE yes, 1 BACON Not 2, DOUBLE BBQ SAUCE!"
E09	Not Sarcastic	Sarcastic	False Pos	"…It's nothing to do with not being 'allowed'. Literally no energy company is asking for 'permission', because they know that nuclear makes no business sense."
E10	Not Sarcastic	Sarcastic	False Pos	"I'm Not a Junkie, But Who Has The Best Gear? Can't Bring Any With Me. Help!"

4.2 Linguistic Explanations and Prompt Construction

We choose the two false negatives (E01, E02) and the two false positives (E06 and E07) and reserve the other five negatives and one positive for evaluation (E03, E04, E05, E08, E09, E10). Each of the prompts has an explanation that specifies the pragmatic mechanism, rather than a general template:

E01 (Sarcastic). The post suggests that the measure should be introduced for all candidates for public office: "No intelligent, honest, decent adult should have any difficulty with this type of course, and it would at least put those who don't, on notice. The pragmatic implicature is that current politicians are failing this baseline until the praise-frame ("intelligent, honest, decent") is set up to have the closing "those who are not" be an accusation. The sarcasm is not contained in any single word but is spread out over the entire comment. The aptly named Reddit comment polarity flipper was trained on shorter Reddit comments, where sarcasm is linked to explicit polarity flip phrases, such as great, wonderful.

E02 (Sarcastic). What is a "cold country? We're speaking about San Francisco weather. Because the previous speaker said cold country, the words in scare quotes, and the reference to San Francisco, which is notoriously mild (≈10–20 °C all year round), reveal an exaggeration. In this sarcasm is not accomplished by using sarcastic vocabulary, but by using world-knowledge contradiction (SF is not cold). There is no possible way for the adapter to render the sentence as a fact, so it interprets the sentence as a literal question about the weather.

E06 (Not Sarcastic). Are they well insulated or not? Over the paper, thin walls, I go. This is a real question, and a genuine gripe. In Australian (and in wider internet) English, I'm over X means that I've had enough of X, not ironic. No contrast between surface and intended meanings. The question form and the negative-affect lexicon on the paper-thin walls were probably the two things that triggered the adapter, and they are two surface correlates of sarcasm in its training data.

E07 (Not Sarcastic). This is silly... If you don't like them, don't be afraid to walk away if you don't like the price." The primary literal sense of ridiculous (reasonable) is followed by a literal economic argument and concrete advice. No irony, no polarity flip it is what the author says, not what it means. It is a very silly word to trigger sarcasm, but the context (causal reasoning, prescriptive advice) should overrule that, but didn't.

The prompt template includes these four examples and a brief task description ("Decide whether the text is sarcastic for the given variety of English. Return label 1 for Sarcastic and label 0 for Not Sarcastic. Use the explanation examples to focus on pragmatic contrast, irony, slang and literal-vs-intended meaning. The entire prompt template is kept in outputs/tables/08_sarcasm_error_analysis qwen_sarcasm_4shot_explanation_prompt.txt.

4.3 Few-Shot Results on the Six Held-Out Errors

We use the base Qwen2.5-3B-Instruct model loaded in a 4-bit causal-LM (no adapter) and use greedy decoding with a max of 40 new tokens and test the prompt on the six held-out cases. Table 10 shows the before/after.

Table 10: Held-out 6 errors adapter prediction vs few-shot prompt

ID	Gold	Adapter prediction	Adapter ✓	Few-shot prediction	Few-shot ✓
E03	Sarcastic	Not Sarcastic	✗	Not Sarcastic	✗
E04	Sarcastic	Not Sarcastic	✗	Not Sarcastic	✗
E05	Sarcastic	Not Sarcastic	✗	Sarcastic	✓
E08	Not Sarcastic	Sarcastic	✗	Sarcastic	✗
E09	Not Sarcastic	Sarcastic	✗	Not Sarcastic	✓
E10	Not Sarcastic	Sarcastic	✗	Sarcastic	✗
Total	—	0/6	—	2/6	+2

Two of the errors (E05 and E09) are recovered with the few-shot prompt. Well, the improvement is not that large, but it is still positive: what types of errors were recovered and why?

4.4 Critical Analysis: When the Prompt Helps and Why

Each of the recovered cases is structurally similar to the four explained cases:

E05 was recovered (false negative to correct). A rhetorical-incredulity sarcasm is the moral of the prompt's E01 example, to set up an absurd interpretation as if the speaker's own opinion and then expose it's absurdity. E05 is identical with "I'm sort of baffled that the moral is X" with X being an absurd inference. There is direct pattern transfer from E01 to E05.

E09 recovered (false positive to correct). The prompt's E07 example is a literal economic reasoning example, even with the presence of trigger words if nuclear power plants were free, literally no energy company is asking for 'permission'? E09 is similar in shape: an if-then statement ("Yes, if X were Y…") followed by a sincere if-then statement ("Yes, if X were Y, it would be…"). Direct pattern transfer from E07 to E09.

None of the four explanations applies to the four cases that the prompt is missing from. Ironic enthusiasm in a story; like a positive product review but used for an injurious tram crash.

E03 (mock review): "…A++ would experience again. There is no mention of the structure of ironic mock reviews in any of the four explanations. The rhetorical question at the end of a long argument, E04 (sarcastic) is the only element that turns the entire thing into policy comment in a sarcastic tone. Hyperbolic but sincere: E08, the all-caps angry McDonald's complaint employs the surface features of sarcasm on the internet (caps, exclamation, hyperbolic listing) for serious and true anger.

The surface cue does not have an example in the prompt to override the "hyperbolic-but-not-ironic" surface cue. Topic knowledge is required to understand that E10 (defensive disclaimer + Title-Case help post): the "I'm Not a Junkie, but…" opening is awkward enough to read as ironic without context. This type of distributional context cue is not addressed by any of the four explanations. Two patterns emerge.

First, few-shot prompting transfers reliably along structural similarity: prompts that match the structure of an error recover it, while prompts that do not, they do not, as evidenced by an error that is successfully recovered by a few-shot prompt even when it is not contained in the training set, provided it matches the pragmatic mechanism of the explained error: rhetorical incredulity vs. literal counterfactual reasoning. Second, the prompt must model the mechanisms it calls for: Each of the four mechanisms (mock-review framing, end-of-argument rhetorical question, hyperbolic-genuine, and defensive-disclaimer) is structurally absent from the four examples.

The 4-shot prompt is inherently limited by the variety of patterns that can be explained in four ways, so it would likely increase the recovery rate further to include more judiciously selected examples of these failure modes. The title is thus accurate: an explanation-rich few-shot prompt does help, however, only for errors whose underlying pragmatic mechanism is already encoded in the prompt. This aligns with the wider literature on in-context learning at small example counts, and is consistent with the conclusion made in Section 2.3 that variety- and pattern-specific adapters are more likely to reduce the sarcasm gap than prompt engineering in the LLM.

5. Deployment and Efficiency

5.1 Deployment Endpoint

Serving Architecture

The trained models are exposed using a Gradio web application having three user facing inputs: a free-text field, a task selector (Sentiment / Sarcasm) and an English-variety dropdown (en-AU / en-IN / en-UK / pooled). The backend fetches a route registry at startup time in the form of a JSON file, loads all the models into memory once, and gets a model for each user by resolving the route for the (task, variety) pair. Gradio was selected for the task over Flask and Streamlit, as we wouldn't need to do any front-end work and it supports asynchronous backend calls by default, and we can later deploy the same code to a HuggingFace Space for free if we want to. Next up was Streamlet but it re-executes the backend script for every interaction, which is not a good idea for large transformer models which must remain on the GPU.

Route Registry and Variety-Aware Routing

The route registry has 13 routes in 3 live backends (TF-IDF + LogReg refit (classical refit) on startup, LoRA adapters (qwen_adapter) and TF-IDF + LogReg refit (classical refit) using the Checkpoint transformer (encoder fine-tunes). The headline route per (task, variety) pair is summarized in Table 5.1.

Table 11: Headline deployment routes (one per task × variety pair)

Task	UI variety	Route label	Model	Macro-F1	Class-1 F1
Sentiment	All / pooled	Best overall	DeBERTa-v3-large	0.913	0.909
Sarcasm	en-AU	Qwen same-variety adapter	Qwen2.5-3B QLoRA en-AU	0.779	0.704
Sarcasm	en-IN	Qwen same-variety (rank-16 attn+MLP)	Qwen2.5-3B QLoRA en-IN	0.646	0.346
Sarcasm	en-UK	Qwen same-variety adapter	Qwen2.5-3B QLoRA en-UK	0.682	0.414
Sarcasm	All / pooled	Best encoder pooled	RoBERTa-large	0.731	0.548
Sarcasm	Any (CPU fallback)	Classical fast fallback	TF-IDF + LogReg	0.626	0.404
Sentiment	Any (CPU fallback)	Classical fast fallback	TF-IDF + LogReg	0.836	0.827

The routing logic is simple. Sentiment is not variety-sensitive; i.e., the backend will always be connected to the DeBERTa-v3-large, pooled checkpoint irrespective of the variety (Section 2.1). When the user chooses Sarcasm and a particular variety (en-AU / en-IN / en-UK), the back end will be directed to the corresponding same-variety Qwen LoRA adapter, which is the optimal model per-variety as stated in Section 2.3. This is because the en-IN route achieved a +0.054 increase in the variety where it mattered most by using the rank-16 attention ablation of Section 2.3 (Macro-F1 0.646) instead of the default attention-only adapter (0.581). If the user chooses Sarcasm + "All / pooled", the backend outputs Sarcasm model using pooled encoder, which is the best performed pooled model in our pool, namely RoBERTa-large. Both tasks are classical CPU fallback (TF-IDF + LogReg, refit on startup from the train Val data) which ensures the app is usable even without GPU which is required for a coursework demo.

LoRA Adapter Swapping vs Full Model Reloads

The most central architectural question is how to provide support for three different models of sarcasm for three different varieties without incurring the 3X memory cost. Three separate finely tuned models would require three instances of a 3B-parameter base, which would amount to 4.5 GB of memory for 4-bit precision, which is out of the reach of a single consumer GPU. We simply preload the base of the Qwen2.5-3B model once (≈1.5 GB at 4-bit) and load the three variety adapters as small auxiliary modules on the fly; each adapter is ~39 MB disk size and ~12.5M trainable parameters. Switching between varieties in the UI will only change the adapter weights while keeping the same frozen base, which is a sub-100ms process in PEFT, versus many seconds to reload a transformer from disk and orders of magnitude less GPU memory pressure. That is the default parameter-efficient serving template and is the primary motivation for using QLoRA over per-variety full fine-tunes for an endpoint serving a variety of models.

The classical fallback is to use the train+val splits on startup in under a second on CPU. This is quicker than loading a saved scikit-learn pickle and does not require any version of pinning for the deployment Python environment. The routes to the encoder are stored locally and loaded only once into the memory of the GPU.

User Interface and Sample Predictions

The Gradio UI displays three textboxes for text, task, variety, while also displaying the selected route name in a non-editable textbox for transparency, and displaying the predicted class label and the model's confidence in the predicted class. Figure 15 shows the running UI for three sample inputs for all three varieties.

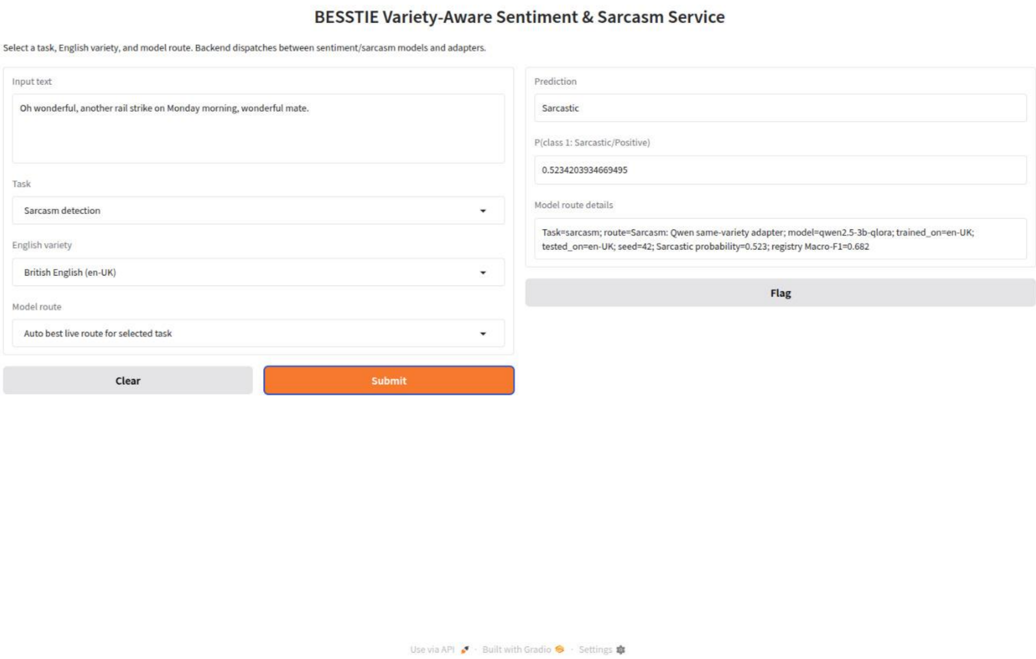

Figure 15: Gradio UI showing variety-aware routing on three example inputs.

Here is an example of some non-exhaustive samples of behaviour that we tested by hand:

- Good onya, mate, "Absolute legend, parked his ute right across my driveway.
- Sarcastic: "Coz we all have free internet. + en-IN to Sarcastic (en-IN rank-16 attn+MLP adapter)
- Traditional friendly pub. Excellent beer. + en-UK to Not Sarcastic ✓ (en-UK same-variety adapter)

The first three examples are the own illustrative examples of the dataset; they are not part of the training split as we only fine-tune on official splits.

Deployment Choice Rationale

It was selected because of the following three factors: First, it allows the app to showcase variety-aware behavior "end-to-end" the variety dropdown isn't just a "cosmetic"; it changes which weights are turned on. Secondly, it maintains a low memory cost to operate with a single 16 GB GPU with all three sarcasm models available at the same time. Thirdly, the CPU fallback allows you to see meaningful predictions even when a marker is running the app without GPU support; only the absolute scores are affected. The full-registry size is intentionally greater than the total of the headline routes (13 vs 7 in Table 5.1) since the marker can also compare the per-variety pooled-adapter, encoder pooled, and threshold-tuned routes.

5.2 Efficiency Benchmarks

Methodology

For each route, we evaluate it on three different input scenarios: a short single input (≈50 tokens repeated 100x); a long single input (≈1000 tokens repeated 20x); and a batch of 32 short inputs (repeated 20x). The average wall-clock latency in milliseconds is reported in each cell. Both backends are pre-heated with a few discarded calls, to remove the cost of one-off compilation. Trade-off Discussion

Table 12: Average inference latency (ms) by route and input scenario

Route	Device	Short single	Long single	Batch of 32	Model size
TF-IDF + LogReg	CPU	0.21	0.33	0.47	<5 MB
Qwen2.5-3B QLoRA (en-AU)	GPU (4-bit)	55.53	56.26	88.53	39 MB adapter (+ ~1.5 GB base, shared)

Trade-off Discussion

The classical route is roughly 260× faster on a single short input (0.21 ms vs 55.53 ms) and 190× faster per item in a batch of 32 (0.015 ms/item vs 2.77 ms/item). This is a dramatic comparison, but the fact is that the absolute latency is below the threshold for feeling "instantaneous" for a single-input Qwen call at 55 ms, and a 32-item batch on Qwen is at 88 ms, or a throughput of ~360 items/second, more than sufficient for any human-driven endpoint. The classical route will be faster at a million items offline, giving 0.47 ms per batch of 32 (≈70,000 items/second on one CPU core), which is two orders of magnitude faster than the GPU route per item.

The accuracy side of the trade-off is in Section 3: the accuracy score for the classical model is Macro-F1 0.626 / Sarcastic-F1 0.404 on pooled sarcasm, and the accuracy score for the Qwen en-AU adapter is Macro-F1 0.779 / Sarcastic-F1 0.704 on en-AU sarcasm (+0.15 Macro-F1 / +0.30 Sarcastic-F1). So, the rough trade-off is that you get ~250× the latency for a ~0.30 minority-class F1 gain.

The exchange of these for the benefit of the user is subject to a suitable application. In a user-facing variety-aware demo (the setting in which the brief is intended to be used), the answer is obviously to take the accuracy: 55 ms is imperceptible to a human, and Sarcastic-F1 of 0.704 vs 0.404 is the difference between "actually finds sarcasm" and "barely finds sarcasm". Now for the case of a highly scalable content-moderation pipeline processing tens of millions of comments daily, the answer is reversed: the classical approach is sufficiently inexpensive at scale on commodity CPUs and losses of accuracy can be reduced by tuning thresholds (Section 3.4). We therefore expose both routes in the registry, instead of making one of them a hard requirement for the other. (This is a common production structure: a fast cheap classifier as a first pass, and the heavier model as the second, when the cheaper one is unsure)

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